

$\mathcal{K}_{0^+}^{\#1} \quad \mathcal{K}_{0^+}^{\#1}{}_{\alpha\beta\chi}$

$\mathcal{K}_{0^+}^{\#1} \dagger$	$-\frac{\Upsilon_1 k^2}{16}$	0
$\mathcal{K}_{0^+}^{\#1} \dagger^{\alpha\beta\chi}$	0	$\Upsilon_2 k^2$

$\mathcal{K}_{1^+}^{\#1} \quad \mathcal{K}_{1^+}^{\#2}{}_{\alpha\beta} \quad \mathcal{K}_{1^+}^{\#1}{}_{\alpha} \quad \mathcal{K}_{1^+}^{\#2}{}_{\alpha}$

$\mathcal{K}_{1^+}^{\#1} \dagger^{\alpha\beta}$	$-\frac{\Upsilon_3 k^2}{16}$	$\frac{\Upsilon_3 k^2}{8\sqrt{2}}$	0	0
$\mathcal{K}_{1^+}^{\#2} \dagger^{\alpha\beta}$	$\frac{\Upsilon_3 k^2}{8\sqrt{2}}$	$-\frac{\Upsilon_3 k^2}{8}$	0	0
$\mathcal{K}_{1^+}^{\#1} \dagger^{\alpha}$	0	0	0	0
$\mathcal{K}_{1^+}^{\#2} \dagger^{\alpha}$	0	0	0	0

$\mathcal{K}_{2^+}^{\#1}{}_{\alpha\beta} \quad \mathcal{K}_{2^+}^{\#1}{}_{\alpha\beta\chi}$

$\mathcal{K}_{2^+}^{\#1} \dagger^{\alpha\beta}$	$\frac{\Upsilon_4 k^2}{16}$	0
$\mathcal{K}_{2^+}^{\#1} \dagger^{\alpha\beta\chi}$	0	$\frac{\Upsilon_5 k^2}{2}$

$\mathcal{J}_{0^+}^{\#1} \quad \mathcal{J}_{0^+}^{\#1}{}_{\alpha\beta\chi}$

$\mathcal{J}_{0^+}^{\#1} \dagger$	$\frac{1}{\text{Det}(0^+)}$	0
$\mathcal{J}_{0^+}^{\#1} \dagger^{\alpha\beta\chi}$	0	$\frac{1}{k^2 \binom{(4)}{\kappa_{15}} \binom{(4)}{\kappa_{16}}}$

$\mathcal{J}_{1^+}^{\#1}{}_{\alpha\beta} \quad \mathcal{J}_{1^+}^{\#2}{}_{\alpha\beta} \quad \mathcal{J}_{1^+}^{\#1}{}_{\alpha} \quad \mathcal{J}_{1^+}^{\#2}{}_{\alpha}$

$\mathcal{J}_{1^+}^{\#1} \dagger^{\alpha\beta}$	$\frac{1}{3 \text{Det}(1^+)}$	$-\frac{\sqrt{2}}{3 \text{Det}(1^+)}$	0	0
$\mathcal{J}_{1^+}^{\#2} \dagger^{\alpha\beta}$	$-\frac{\sqrt{2}}{3 \text{Det}(1^+)}$	$\frac{2}{3 \text{Det}(1^+)}$	0	0
$\mathcal{J}_{1^+}^{\#1} \dagger^{\alpha}$	0	0	0	0
$\mathcal{J}_{1^+}^{\#2} \dagger^{\alpha}$	0	0	0	0

$\mathcal{J}_{2^+}^{\#1}{}_{\alpha\beta} \quad \mathcal{J}_{2^+}^{\#1}{}_{\alpha\beta\chi}$

$\mathcal{J}_{2^+}^{\#1} \dagger^{\alpha\beta}$	$\frac{1}{\text{Det}(2^+)}$	0
$\mathcal{J}_{2^+}^{\#1} \dagger^{\alpha\beta\chi}$	0	$\frac{1}{\text{Det}(2^')}$

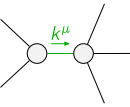
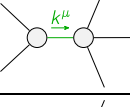
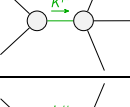
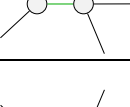
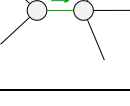
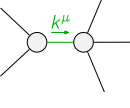
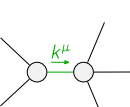
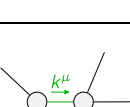
Abbreviations used in matrices

$\Upsilon_1 == 24 \binom{(4)}{\kappa_1} - 10 \binom{(4)}{\kappa_{15}} + 13 \binom{(4)}{\kappa_{16}} + 18 \binom{(4)}{\kappa_3}$
&&
 $\Upsilon_2 == \binom{(4)}{\kappa_{15}} - \binom{(4)}{\kappa_{16}}$
&&
 $\Upsilon_3 == 2 \binom{(4)}{\kappa_{15}} - \binom{(4)}{\kappa_{16}} - 2 \binom{(4)}{\kappa_3}$
&&
 $\Upsilon_4 == 34 \binom{(4)}{\kappa_{15}} - \binom{(4)}{\kappa_{16}} - 18 \binom{(4)}{\kappa_3}$
&&
 $\Upsilon_5 == 2 \binom{(4)}{\kappa_{15}} + \binom{(4)}{\kappa_{16}}$
&&
 $\text{Det}(0^+) == -\frac{1}{16} k^2 (24 \binom{(4)}{\kappa_1} - 10 \binom{(4)}{\kappa_{15}} + 13 \binom{(4)}{\kappa_{16}} + 18 \binom{(4)}{\kappa_3})$
&&

$\text{Det}(1^+) == \frac{3}{16} k^2 (-2 \binom{(4)}{\kappa_{15}} + \binom{(4)}{\kappa_{16}} + 2 \binom{(4)}{\kappa_3})$
&&
 $\text{Det}(2^+) == \frac{1}{16} k^2 (34 \binom{(4)}{\kappa_{15}} - \binom{(4)}{\kappa_{16}} - 18 \binom{(4)}{\kappa_3})$

Lagrangian

$\binom{(4)}{\kappa_1} \partial_\beta \mathcal{K}^\delta{}_{\chi\delta} \partial^\chi \mathcal{K}^\alpha{}_{\alpha}{}^\beta + \binom{(4)}{\kappa_{15}} \partial_\chi \mathcal{K}^\delta{}_{\beta\delta} \partial^\chi \mathcal{K}^\alpha{}_{\alpha}{}^\beta + \frac{1}{2} \binom{(4)}{\kappa_{16}} \partial_\chi \mathcal{K}^\delta{}_{\beta\delta} \partial^\chi \mathcal{K}^\alpha{}_{\alpha}{}^\beta +$
 $\binom{(4)}{\kappa_3} \partial_\beta \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}^\delta{}_{\alpha\chi}{}^\delta + \frac{1}{2} \binom{(4)}{\kappa_{15}} \partial_\alpha \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}^\delta{}_{\beta\chi}{}^\delta + \frac{7}{4} \binom{(4)}{\kappa_{16}} \partial_\alpha \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}^\delta{}_{\beta\chi}{}^\delta -$
 $\frac{1}{2} \binom{(4)}{\kappa_3} \partial_\alpha \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}^\delta{}_{\beta\chi}{}^\delta - \frac{9}{4} \binom{(4)}{\kappa_{15}} \partial_\beta \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}^\delta{}_{\chi\alpha}{}^\delta + \frac{9}{8} \binom{(4)}{\kappa_{16}} \partial_\beta \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}^\delta{}_{\chi\alpha}{}^\delta +$
 $\frac{5}{4} \binom{(4)}{\kappa_3} \partial_\beta \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}^\delta{}_{\chi\alpha}{}^\delta - 2 \binom{(4)}{\kappa_{15}} \partial^\chi \mathcal{K}^\alpha{}_{\alpha}{}^\beta \partial_\delta \mathcal{K}^\delta{}_{\chi\beta}{}^\delta - \binom{(4)}{\kappa_{16}} \partial^\chi \mathcal{K}^\alpha{}_{\alpha}{}^\beta \partial_\delta \mathcal{K}^\delta{}_{\chi\beta}{}^\delta - \frac{5}{4} \binom{(4)}{\kappa_{15}} \partial_\alpha \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}^\delta{}_{\beta\chi}{}^\delta +$
 $\frac{1}{8} \binom{(4)}{\kappa_{16}} \partial_\alpha \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}^\delta{}_{\beta\chi}{}^\delta + \frac{1}{4} \binom{(4)}{\kappa_3} \partial_\alpha \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}^\delta{}_{\beta\chi}{}^\delta + \binom{(4)}{\kappa_{15}} \partial_\delta \mathcal{K}^{\alpha\beta\chi} \partial^\delta \mathcal{K}^{\alpha\beta\chi} + \binom{(4)}{\kappa_{16}} \partial_\delta \mathcal{K}^{\alpha\beta\chi} \partial^\delta \mathcal{K}^{\alpha\beta\chi}$

Added source term(s):	$\mathcal{K}^{\alpha\beta\chi} \mathcal{J}_{\alpha\beta\chi}$		
Source constraint(s)	# constraint(s)	Covariant form	
$\mathcal{J}_{1^+}^{\#2\alpha} == 0$	3	$\partial_\chi \partial_\beta \mathcal{J}^{\beta\alpha\chi} == 0$	
$\mathcal{J}_{1^+}^{\#1\alpha} == 0$	3	$\partial_\chi \partial^\alpha \mathcal{J}^\beta{}_\beta{}^\chi + \partial_\chi \partial^\chi \mathcal{J}^{\beta\alpha}{}_\beta == \partial_\chi \partial_\beta \mathcal{J}^{\beta\alpha\chi}$	
$2 \mathcal{J}_{1^+}^{\#1\alpha\beta} + \mathcal{J}_{1^+}^{\#2\alpha\beta} == 0$	3	$\partial_\chi \mathcal{J}^{\alpha\beta\chi} + \partial_\chi \mathcal{J}^{\chi\alpha\beta} == \partial_\chi \mathcal{J}^{\beta\alpha\chi}$	
Total # constraint(s):	9		
Resolved pole(s)	# polarization(s)	Square mass	Residue
	1	0	$\frac{-50 \binom{(4)}{\kappa_{15}} - 7 \binom{(4)}{\kappa_{16}} + 18 \binom{(4)}{\kappa_3}}{(2 \binom{(4)}{\kappa_{15}} + \binom{(4)}{\kappa_{16}}) (34 \binom{(4)}{\kappa_{15}} - \binom{(4)}{\kappa_{16}} - 18 \binom{(4)}{\kappa_3})}$
	1	0	$\frac{-108 \binom{(4)}{\kappa_{15}}^2 - 52 \binom{(4)}{\kappa_{15}} \binom{(4)}{\kappa_{16}} - 11 \binom{(4)}{\kappa_{16}}^2 + 56 \binom{(4)}{\kappa_{15}} \binom{(4)}{\kappa_3} + 4 \binom{(4)}{\kappa_{16}} \binom{(4)}{\kappa_3} - 12 \binom{(4)}{\kappa_3}^2}{(2 \binom{(4)}{\kappa_{15}} + \binom{(4)}{\kappa_{16}}) (34 \binom{(4)}{\kappa_{15}} - \binom{(4)}{\kappa_{16}} - 18 \binom{(4)}{\kappa_3}) (2 \binom{(4)}{\kappa_{15}} - \binom{(4)}{\kappa_{16}} - 2 \binom{(4)}{\kappa_3})}$
	1	0	$\frac{50 \binom{(4)}{\kappa_{15}} + 7 \binom{(4)}{\kappa_{16}} - 18 \binom{(4)}{\kappa_3}}{(2 \binom{(4)}{\kappa_{15}} + \binom{(4)}{\kappa_{16}}) (34 \binom{(4)}{\kappa_{15}} - \binom{(4)}{\kappa_{16}} - 18 \binom{(4)}{\kappa_3})}$
	1	0	$\frac{-332 \binom{(4)}{\kappa_{15}}^2 + 44 \binom{(4)}{\kappa_{15}} \binom{(4)}{\kappa_{16}} - 3 \binom{(4)}{\kappa_{16}}^2 + 376 \binom{(4)}{\kappa_{15}} \binom{(4)}{\kappa_3} - 28 \binom{(4)}{\kappa_{16}} \binom{(4)}{\kappa_3} - 108 \binom{(4)}{\kappa_3}^2}{(2 \binom{(4)}{\kappa_{15}} + \binom{(4)}{\kappa_{16}}) (34 \binom{(4)}{\kappa_{15}} - \binom{(4)}{\kappa_{16}} - 18 \binom{(4)}{\kappa_3}) (2 \binom{(4)}{\kappa_{15}} - \binom{(4)}{\kappa_{16}} - 2 \binom{(4)}{\kappa_3})}$
	1	0	$\frac{-((-32 \binom{(4)}{\kappa_{15}}^2 + 8 \binom{(4)}{\kappa_{16}}^2 + 32 \binom{(4)}{\kappa_{15}} \binom{(4)}{\kappa_3} + 16 \binom{(4)}{\kappa_{16}} \binom{(4)}{\kappa_3} + \sqrt{((-2 \binom{(4)}{\kappa_{15}} + \binom{(4)}{\kappa_{16}} + 2 \binom{(4)}{\kappa_3})^2 (1924 \binom{(4)}{\kappa_{15}}^2 + 700 \binom{(4)}{\kappa_{15}} \binom{(4)}{\kappa_{16}} + 193 \binom{(4)}{\kappa_{16}}^2 - 1224 \binom{(4)}{\kappa_{15}} \binom{(4)}{\kappa_3} + 36 \binom{(4)}{\kappa_{16}} \binom{(4)}{\kappa_3} + 324 \binom{(4)}{\kappa_3}^2))})}{((2 \binom{(4)}{\kappa_{15}} + \binom{(4)}{\kappa_{16}}) (34 \binom{(4)}{\kappa_{15}} - \binom{(4)}{\kappa_{16}} - 18 \binom{(4)}{\kappa_3}) (2 \binom{(4)}{\kappa_{15}} - \binom{(4)}{\kappa_{16}} - 2 \binom{(4)}{\kappa_3}))}$
	1	0	$\frac{(32 \binom{(4)}{\kappa_{15}}^2 - 8 \binom{(4)}{\kappa_{16}}^2 - 32 \binom{(4)}{\kappa_{15}} \binom{(4)}{\kappa_3} - 16 \binom{(4)}{\kappa_{16}} \binom{(4)}{\kappa_3} + \sqrt{((-2 \binom{(4)}{\kappa_{15}} + \binom{(4)}{\kappa_{16}} + 2 \binom{(4)}{\kappa_3})^2 (1924 \binom{(4)}{\kappa_{15}}^2 + 700 \binom{(4)}{\kappa_{15}} \binom{(4)}{\kappa_{16}} + 193 \binom{(4)}{\kappa_{16}}^2 - 1224 \binom{(4)}{\kappa_{15}} \binom{(4)}{\kappa_3} + 36 \binom{(4)}{\kappa_{16}} \binom{(4)}{\kappa_3} + 324 \binom{(4)}{\kappa_3}^2))})}{((2 \binom{(4)}{\kappa_{15}} + \binom{(4)}{\kappa_{16}}) (34 \binom{(4)}{\kappa_{15}} - \binom{(4)}{\kappa_{16}} - 18 \binom{(4)}{\kappa_3}) (2 \binom{(4)}{\kappa_{15}} - \binom{(4)}{\kappa_{16}} - 2 \binom{(4)}{\kappa_3}))}$
	1	0	$\frac{(272 \binom{(4)}{\kappa_{15}}^2 + 4 \binom{(4)}{\kappa_{16}}^2 + 80 \binom{(4)}{\kappa_{16}} \binom{(4)}{\kappa_3} + 144 \binom{(4)}{\kappa_3}^2 - 16 \binom{(4)}{\kappa_{15}} (9 \binom{(4)}{\kappa_{16}} + 26 \binom{(4)}{\kappa_3}) - \sqrt{((-34 \binom{(4)}{\kappa_{15}} + \binom{(4)}{\kappa_{16}} + 18 \binom{(4)}{\kappa_3})^2 (452 \binom{(4)}{\kappa_{15}}^2 + 124 \binom{(4)}{\kappa_{15}} \binom{(4)}{\kappa_{16}} + 81 \binom{(4)}{\kappa_{16}}^2 - 328 \binom{(4)}{\kappa_{15}} \binom{(4)}{\kappa_3} + 100 \binom{(4)}{\kappa_{16}} \binom{(4)}{\kappa_3} + 132 \binom{(4)}{\kappa_3}^2))})}{((2 \binom{(4)}{\kappa_{15}} + \binom{(4)}{\kappa_{16}}) (34 \binom{(4)}{\kappa_{15}} - \binom{(4)}{\kappa_{16}} - 18 \binom{(4)}{\kappa_3}) (2 \binom{(4)}{\kappa_{15}} - \binom{(4)}{\kappa_{16}} - 2 \binom{(4)}{\kappa_3}))}$
	1	0	$\frac{(272 \binom{(4)}{\kappa_{15}}^2 + 4 \binom{(4)}{\kappa_{16}}^2 + 80 \binom{(4)}{\kappa_{16}} \binom{(4)}{\kappa_3} + 144 \binom{(4)}{\kappa_3}^2 - 16 \binom{(4)}{\kappa_{15}} (9 \binom{(4)}{\kappa_{16}} + 26 \binom{(4)}{\kappa_3}) + \sqrt{((-34 \binom{(4)}{\kappa_{15}} + \binom{(4)}{\kappa_{16}} + 18 \binom{(4)}{\kappa_3})^2 (452 \binom{(4)}{\kappa_{15}}^2 + 124 \binom{(4)}{\kappa_{15}} \binom{(4)}{\kappa_{16}} + 81 \binom{(4)}{\kappa_{16}}^2 - 328 \binom{(4)}{\kappa_{15}} \binom{(4)}{\kappa_3} + 100 \binom{(4)}{\kappa_{16}} \binom{(4)}{\kappa_3} + 132 \binom{(4)}{\kappa_3}^2))})}{((2 \binom{(4)}{\kappa_{15}} + \binom{(4)}{\kappa_{16}}) (34 \binom{(4)}{\kappa_{15}} - \binom{(4)}{\kappa_{16}} - 18 \binom{(4)}{\kappa_3}) (2 \binom{(4)}{\kappa_{15}} - \binom{(4)}{\kappa_{16}} - 2 \binom{(4)}{\kappa_3}))}$
Resolved unitarity condition(s):	(Demonstrably impossible)		